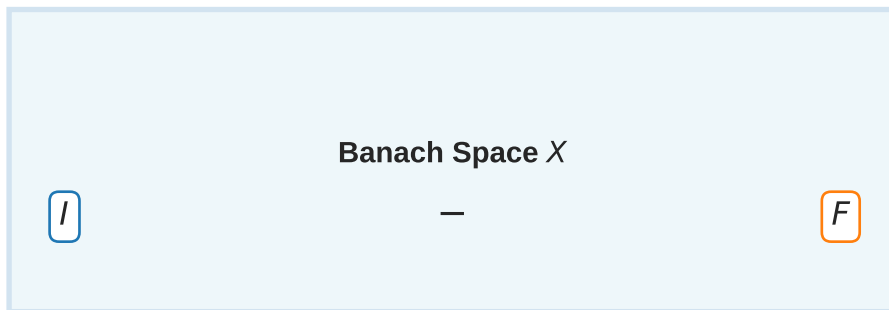


Leray-Schauder Construction

Leray-Schauder Degree

For map: $\Phi = I - F$ where F is compact



Maps to finite dimensional subspace

for degree computation

Key Properties

1. Existence of Solutions

If $\deg(I - F, \Omega, y) \neq 0$, then

$$\exists x \in \Omega : x - Fx = y$$

2. Homotopy Invariance

Degree is invariant under continuous deformations

that don't change boundary behavior

3. Infinite Dimensional Extension

Extends Brouwer degree to Banach spaces

with compact perturbations